BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Hardware and Topologies

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What distinguishes a WAN?

2. Why use shared computing standards?
   A) Every device needs one maker
   B) Different products work together
   C) No communication rules are needed
   D) All hardware becomes identical

3. "SMTP's role is synchronising folders and read status across mail clients." Correct this. Explain why.

4. A star LAN has one central switch. "One broken workstation cable stops the whole network." Correct this. Compare a broken cable with a failed switch.

Today's objectives

Explain the core ideas of network hardware and topologies using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

NIC
Provides a device's connection to a network.

Switch
Forwards frames between devices within a LAN using MAC address information.

Router
Connects networks and forwards packets using IP address/routing information.

Star and mesh
Star connects devices via a central point; mesh offers multiple potential paths between nodes.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

NIC
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Star and mesh
Star connects devices via a central point; mesh offers multiple potential paths between nodes.

I Do - Worked model

Explain how data travels from a classroom PC to a server on another network.

- The PC connects via its NIC.
- A switch forwards traffic inside the LAN.
- A router forwards the packet to the other network.
- The destination network delivers data to the server.

We Do - Guided practice

1. Compare star and mesh topology after one connection fails.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the number of links in a fully connected mesh of five computers.

```
devices = 5
links = devices * (devices - 1) DIV 2
print(links)
```

You Do - Independent exam questions

Q2. State what a switch does in a LAN. [2 marks]

Q3. Give the purpose of a router. [2 marks]

Q4. Calculate full-mesh connections for four nodes. [2 marks]

Exit Ticket - True or False?

A switch normally forwards using IP routes between networks.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A WAN links networks over a large area, often using links provided by other organisations.

2. B - Different products work together

3. Sending email from a client to a mail server and between mail servers.

4. A broken workstation cable usually affects just that device. If the central switch fails, devices using it lose contact. Available paths determine which connections still work.

I Do - Explain the model

The PC connects via its NIC.

A switch forwards traffic inside the LAN.

A router forwards the packet to the other network.

The destination network delivers data to the server.

We Do - Model answers

1. Explain the trade-off: mesh uses more connections and costs more.

Code First - Answer

```
devices = 5
links = devices * (devices - 1) DIV 2
print(links)
```

Expected output: 10

You Do - Self-mark

Q2: Forwards frames to the appropriate device/port using learned MAC address information.

Q3: Connects separate networks and forwards packets between them using IP routing.

Q4: 4 x 3 / 2=6 connections.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.